

Trainings ▾

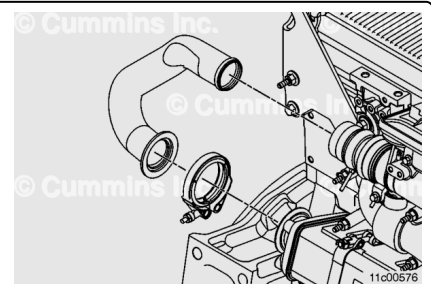
Note: A video supports the information provided in this procedure. Read the procedure prior to viewing the video.

Test

Automotive with CM870

⚠ WARNING ⚠

The exhaust and exhaust components can remain hot after the engine has been shut down or secured. To avoid the risk of fire, property damage, burns or other serious personal injury, allow the exhaust system to cool before beginning this procedure or repair and make sure that no combustible materials are located where they might come in contact with hot exhaust or exhaust components.



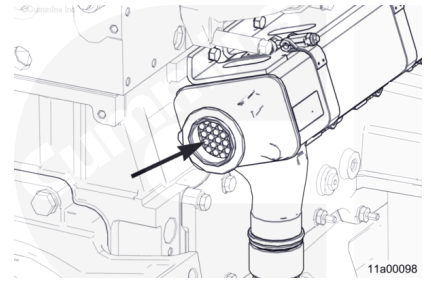
Note : Test is performed with exhaust gas recirculation (EGR) cooler installed on the engine with Fleetguard® Coolant Test Strip CC2602B.

Coolant Test Strip - EGR Cooler Exhaust Outlet

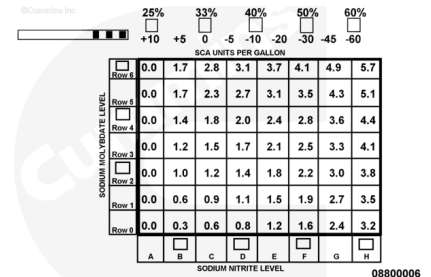
- Test Fleetguard® 3-Way Coolant Test strip, following instructions on the package, with distilled water to create a baseline for analysis.
- Remove rear EGR connection tube. Refer to Procedure 011-025 in Section 11.
(</qs3/pubsys2/xml/en/procedures/10/10-011-025-tr.html>)

- Remove a small sample of soot from the EGR cooler exhaust outlet.
- Place soot sample into a small clean container.

- Mix soot sample with 25 ml (0.8 oz) of distilled water.



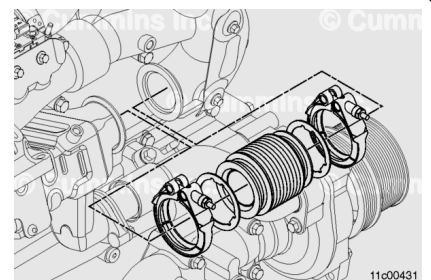
- Test soot and water solution with a Fleetguard® 3-Way Coolant Test strip by following the instructions on the package.
- EGR cooler is reusable if all the pads on the test strip keep their baseline color.
- Install exhaust outlet tube and return to published troubleshooting.
- A change in any of the three test pads indicates coolant is in the exhaust outlet side of the cooler. Proceed to inlet side test.
- **Note :** It may be necessary to test raw coolant from the cooling system to make sure the Fleetguard® 3-Way Coolant Test strips will react if the type of coolant is unknown.



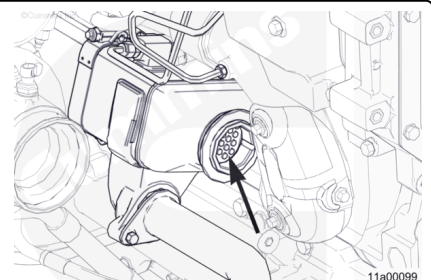
Note : Test is performed with the EGR cooler installed on the engine with Fleetguard® Coolant Test Strip CC2602B.

Coolant Test Strip - EGR Cooler Exhaust Inlet

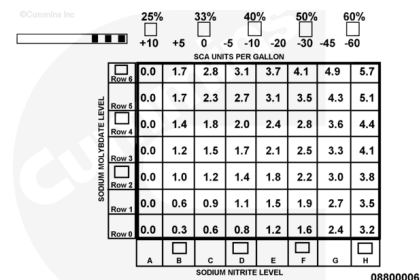
- Test Fleetguard® 3-Way Coolant Test Strip, following instructions on the package, with distilled water to create a baseline for analysis.
- Remove front EGR cooler connection. Refer to Procedure 011-024 in Section 10.
(/qs3/pubsys2/xml/en/procedures/10/10-011-024.html)



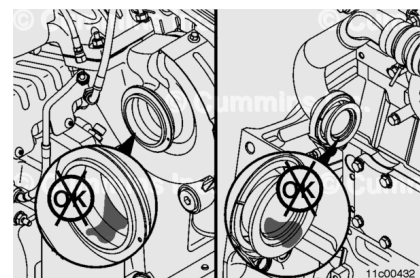
- Remove a small sample of soot from the EGR cooler exhaust inlet.
- Place soot sample into a small clean container.
- Mix soot sample with 25 ml (0.8 oz) of distilled water.



- Test soot and water solution with a Fleetguard® 3-Way Coolant Test strip by following the instructions on the package.
- A change in any of the three test pads indicates coolant is in the exhaust inlet side of the cooler. Proceed to Preparatory Steps.
- EGR cooler is failed if coolant is in the outlet side and **not** in the inlet side of the EGR cooler.



Coolant Test Strip Results Guide		
Coolant in Outlet	Coolant in Inlet	Outcome
Yes	Yes	Pressure test EGR cooler.
No	N/A	EGR cooler is not failed. Return to published troubleshooting.
Yes	No	EGR cooler is failed. Replace cooler.
Yes	Yes	Pressure test EGR cooler.



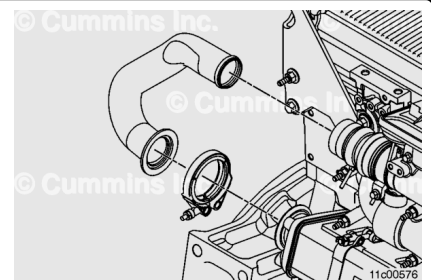
- Inspect turbocharger turbine housing and rear EGR connection tube for any signs of coolant deposits if the EGR cooler is **not** reusable.
- Coolant deposits can be identified as dried white deposits coating the inside of the turbine housing.
- If coolant deposits are found in the turbocharger turbine housing, inspect the turbocharger, EGR valve, EGR mass measurement flow assembly, exhaust transfer connection elbow, EGR crossover tube, and the crankcase pressure sensor for progressive damage.
- Clean the aftertreatment fuel injector, regardless if coolant deposits are present or **not**. Refer to Procedure 011-042 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-042-tr.html)
- Reference the following procedures:
 - Turbocharger - Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
 - EGR valve - Refer to Procedure 011-022 in Section 11. (/qs3/pubsys2/xml/en/procedures/10/10-011-022-tr.html)

- Exhaust transfer connection tube - Refer to Procedure 011-024 in Section 11.
(/qs3/pubsys2/xml/en/procedures/10/10-011-024.html)
- EGR connection tube - Refer to Procedure 011-025 in Section 11.
(/qs3/pubsys2/xml/en/procedures/10/10-011-025-tr.html)
- Crankcase pressure sensor: Refer to Procedure 019-445 in Section 19.
(/qs3/pubsys2/xml/en/procedures/00/00-019-445.html)

Automotive With CM871

⚠ WARNING ⚠

The exhaust and exhaust components can remain hot after the engine has been shut down or secured. To avoid the risk of fire, property damage, burns or other serious personal injury, allow the exhaust system to cool before beginning this procedure or repair and make sure that no combustible materials are located where they might come in contact with hot exhaust or exhaust components.



Note : Test is performed with EGR cooler installed on engine using Fleetguard® 3-Way Coolant Test Kit CC2602B.

Note : It may be necessary to test raw coolant from the cooling system to make sure the Fleetguard® 3-Way Coolant Test strips will react if the type of coolant is unknown.

Coolant Test Strip - EGR Cooler Exhaust Outlet

- Test Fleetguard® 3-Way Coolant Test strip, following instructions on the package, using distilled water to create a baseline for analysis.
- Remove rear EGR connection tube. Refer to Procedure 011-025 in Section 11
(/qs3/pubsys2/xml/en/procedures/10/10-011-025-tr.html).

- Remove a small sample of soot from the EGR cooler exhaust outlet.
- Place soot sample into a small clean container.

- 
- 11a00098

-
- 0.0 25% 33% 40% 50% 60%
- +10 +5 0 -5 -10 -20 -30 -45 -60
- SCA UNITS PER GALLON
- SODIUM NITRATE LEVEL
- | | | | | | | | | |
|-------|-----|-----|-----|-----|-----|-----|-----|-----|
| Row 6 | 0.0 | 1.7 | 2.8 | 3.1 | 3.7 | 4.1 | 4.9 | 5.7 |
| Row 5 | 0.0 | 1.7 | 2.3 | 2.7 | 3.1 | 3.5 | 4.3 | 5.1 |
| Row 4 | 0.0 | 1.4 | 1.8 | 2.0 | 2.4 | 2.8 | 3.6 | 4.4 |
| Row 3 | 0.0 | 1.2 | 1.5 | 1.7 | 2.1 | 2.5 | 3.3 | 4.1 |
| Row 2 | 0.0 | 1.0 | 1.2 | 1.4 | 1.8 | 2.2 | 3.0 | 3.8 |
| Row 1 | 0.0 | 0.6 | 0.9 | 1.1 | 1.5 | 1.9 | 2.7 | 3.5 |
| Row 0 | 0.0 | 0.3 | 0.6 | 0.8 | 1.2 | 1.6 | 2.4 | 3.2 |
| | A | B | C | D | E | F | G | H |
- SODIUM NITRATE LEVEL
- 08800006

Automotive with CM870

Note : If the EGR cooler is being replaced due to an internal coolant leak, the EGR differential pressure tubes must be replaced. Refer to Procedure 011-026 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-026.html>)

Note : Brush away all loose dirt from around the area of the air-handling connections to reduce the possibility of contamination of the interior of the engine.

Note : Use protective caps from the ISX Clean Care Kit, Part Number 4919403, and the original equipment manufacturer (OEM) Clean Care Kit, Part Number 4919425, to cover open connection points.

Note : If the EGR cooler is being replaced due to a malfunction that caused an internal coolant leak and Fault Code 555 and/or 556 is present, the crankcase breather system must be inspected. Maintenance-free crankcase breather inserts must be cleaned and inspected for reuse. Crankcase breather elements with filtration media must be replaced. Refer to Procedure 003-019 in Section 3.

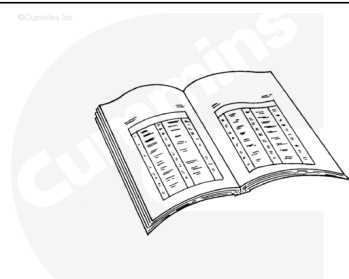
(/qs3/pubsys2/xml/en/procedures/10/10-003-019.html)

- Drain the engine coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
- Remove the air intake pipe to the turbocharger. See equipment manufacturer service information.
- Remove the charge air cooler pipe between the turbocharger and charge air cooler. See equipment manufacturer service information.
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
- Remove the exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11. (/qs3/pubsys2/xml/en/procedures/10/10-011-027-tr.html)
- Remove the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11. (/qs3/pubsys2/xml/en/procedures/10/10-011-031-tr.html)

Automotive With CM871

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : Brush away all loose dirt from around the area of the air handling connections to reduce the possibility of contamination of the interior of the engine.

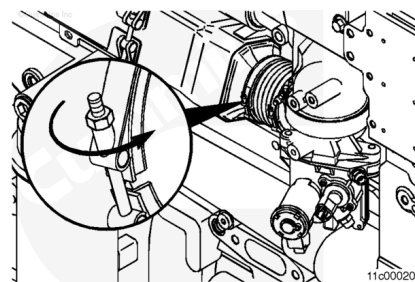
Note : Use protective caps from the ISX Clean Care Kit, Part Number 4919403, and the OEM Clean Care Kit, Part Number 4919425, to cover open connection points.

- Drain the engine coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>)
- Remove the air intake pipe to the turbocharger. See equipment manufacturer service information.
- Remove the charge air cooler pipe between the turbocharger and charge air cooler. See equipment manufacturer service information.
- Remove the aftertreatment adapter pipe and aftertreatment injector lines. Refer to Procedure 011-051 in Section 11.
(</qs3/pubsys2/xml/en/procedures/101/101-011-051-tr.html>) Refer to Procedure 011-043 in Section 11.
(</qs3/pubsys2/xml/en/procedures/101/101-011-043-tr.html>)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html>)
- Remove the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11.
(</qs3/pubsys2/xml/en/procedures/10/10-011-031-tr.html>)

Remove

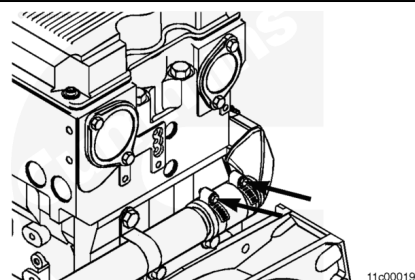
Automotive with CM870

Loosen the V-band clamp between the EGR cooler and the EGR cooler connection. When removing the V-band clamp, do **not** damage the EGR connection tube.



Loosen hose clamps at the rear EGR connection tube.

Remove EGR hose.



⚠ CAUTION ⚠

The straps on the EGR cooler are not serviceable. Adjusting or removing the straps can damage the EGR cooler.

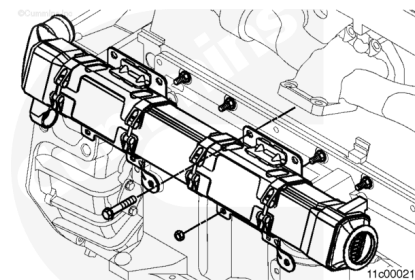
Remove the four capscrews that attach the EGR cooler to the block.

Remove the four nuts that attach the EGR cooler to the lubricating oil cooler assembly.

Remove EGR cooler assembly from the engine.

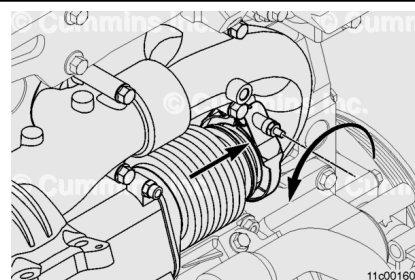
Remove coolant supply and vent jumper tubes if they remain in the block after the cooler is removed.

Note : Use protective caps from the ISX Clean Care Kit, Part Number 4919403, to cover open connection points on both the EGR cooler and EGR plumbing.



Automotive With CM871

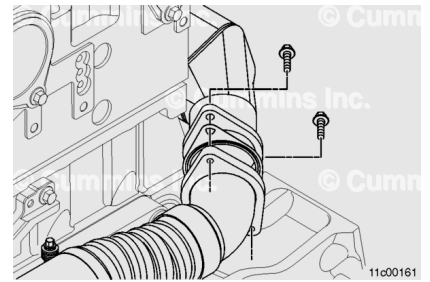
Loosen V-band clamp between the EGR cooler and the exhaust manifold. To prevent damage to the EGR cooler bellows, place the V-band clamp over the exhaust manifold flange.



Loosen rear EGR connection tube retaining capscrews at the EGR cooler gas outlet.

Remove and discard the gasket.

Use protective caps from the ISX Clean Care Kit, Part Number 4919403, to cover open connection points on both the EGR cooler and EGR plumbing.



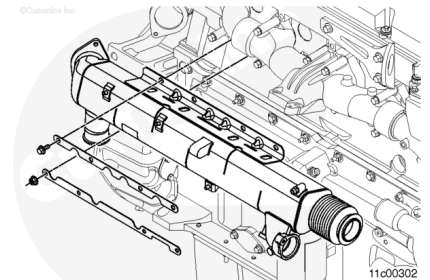
Remove the four capscrews that attach the EGR cooler to the block.

Remove the four nuts that attach the EGR cooler to the lubricating oil cooler assembly.

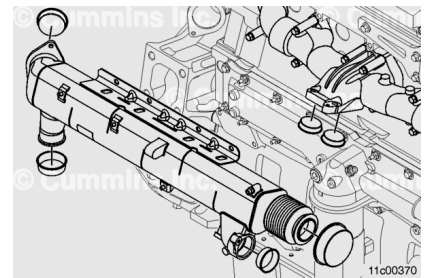
Remove EGR cooler bracket clamping plates from the EGR cooler.

Note : Some EGR coolers use washers instead of clamping plates.

Remove EGR cooler assembly from the engine.



Use protective caps from the ISX Clean Care Kit, Part Number 4919403, to cover open connection points on the EGR cooler and exhaust manifold.

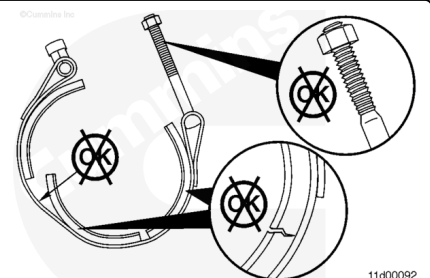


Clean and Inspect for Reuse

Automotive with CM870

Inspect V-band clamp threads for damage. Replace the clamp if damage is found.

Inspect hose clamps for damage. Replace if damage is found.

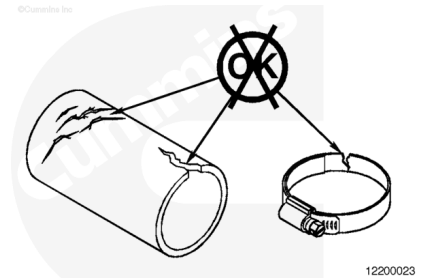


Inspect hose clamp(s) for cracks or other damage.

Replace hose clamp(s) if cracks or other damage is found.

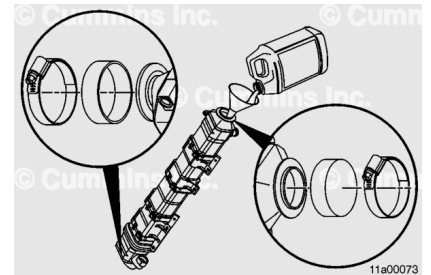
Inspect hose(s) for cracks or other damage.

Replace hose(s) if cracks or other damage is found.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



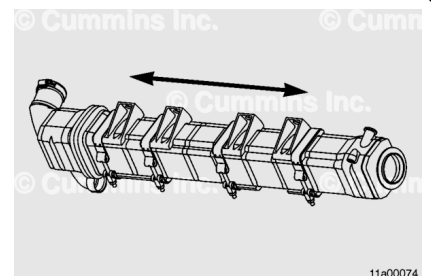
Cap outlet end of EGR cooler with parts supplied in EGR Cooler Leak Check Kit, Part Number 3164225.

Completely fill the EGR cooler with safety solvent.

Cap the gas inlet end of the EGR cooler with parts supplied in the EGR Cooler Leak Test Kit, Part Number 3164225.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Lay EGR cooler on its side.

Allow the EGR cooler to soak for approximately 20 minutes.

Drain approximately 25 percent of the liquid from the cooler and recap.

Properly dispose of the liquid.

Shake the cooler by hand, from end-to-end, for approximately 30 seconds.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

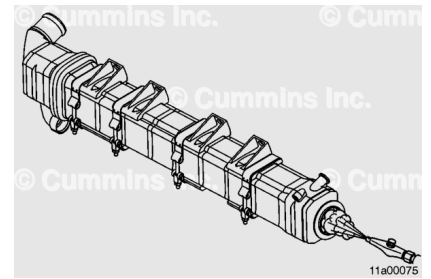
possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not use chlorinated water to clean or rinse the EGR cooler. Chlorides are very corrosive to the stainless alloys internal and external to the EGR cooler.



Drain the remaining safety solvent from the EGR cooler and properly dispose of it.

Use compressed air to dry the inside of the EGR cooler and remove any loose debris or particles.

The EGR cooler can be cleaned again, if necessary.

Cap the outlet end of the EGR cooler with parts supplied in EGR Cooler Leak Test Kit, Part Number 3164225.

Completely fill the EGR cooler with water.

Cap the gas inlet end of the EGR cooler with parts supplied in the EGR Cooler Leak Test Kit, Part Number 3164225.

Shake the cooler by hand, from end-to-end, for approximately 30 seconds.

Drain the water from the EGR cooler and properly dispose of it.

Use compressed air to dry the inside of the EGR cooler.

Clean the EGR valve with safety solvent where the EGR cooler connection attaches to the EGR valve. Refer to Procedure 011-022 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-022-tr.html>)

Inspect the EGR valve where the EGR cooler connection attaches to the EGR valve. Replace the EGR valve if cracks or other damage is found. Refer to Procedure 011-022 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-022-tr.html>)

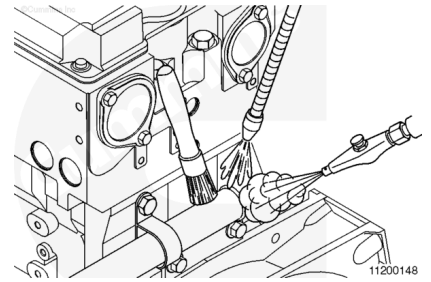
Replace the EGR cooler if cracks or other damage is found.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



With EGR connection tube attached, clean the cooler end of the EGR connection tube with safety solvent.

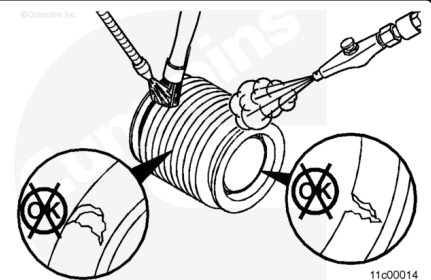
Dry with compressed air.

Inspect EGR connection tube for cracks or other damage, while still attached to the block. Replace the EGR connection tube, if necessary. Refer to Procedure 011-025 in Section 11.

(/qs3/pubsys2/xml/en/procedures/10/10-011-025-tr.html)

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⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

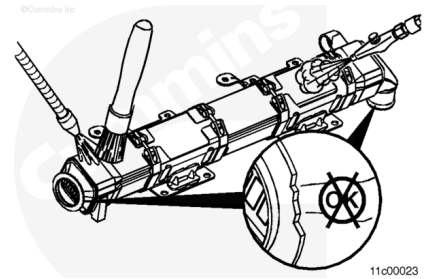
Clean EGR cooler connection with safety solvent and dry with compressed air.

Inspect EGR cooler connection for cracks or other damage. Replace EGR cooler connection if cracks or other damage is found. Refer to Procedure 011-024 in Section 11.

(/qs3/pubsys2/xml/en/procedures/10/10-011-024.html)

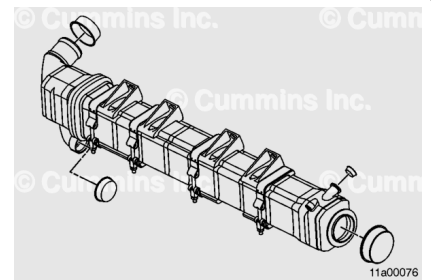
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



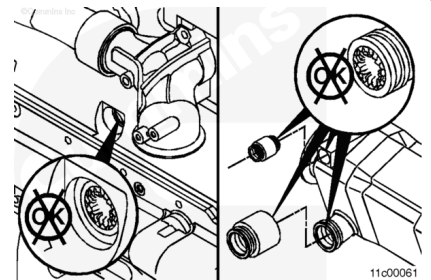
Clean mounting surfaces with safety solvent to remove any coolant deposits or debris.

After parts are clean and dry, cover open connection points with protective caps from the ISX Clean Care Kit, Part Number 4919403, or with heavy tape, if needed.



Clean opening in the block where the EGR cooler coolant supply and vent jumper tubes are installed to remove any coolant deposits or debris. Otherwise, the o-ring joint can leak.

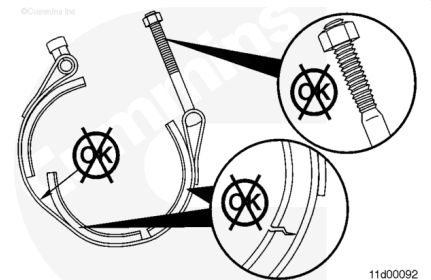
Clean coolant supply and vent jumper tubes to remove any coolant deposits or debris.



Automotive With CM871

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



⚠ WARNING ⚠

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Inspect V-band clamp threads for damage. Replace the clamp if damage is found.

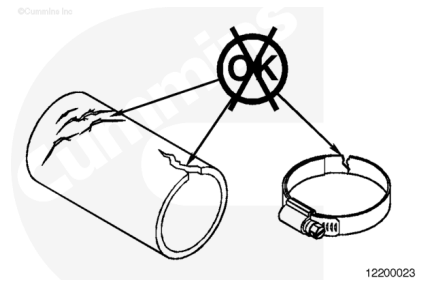
Inspect hose clamps for damage. Replace if damage is found.

Inspect hose clamp(s) for cracks or other damage.

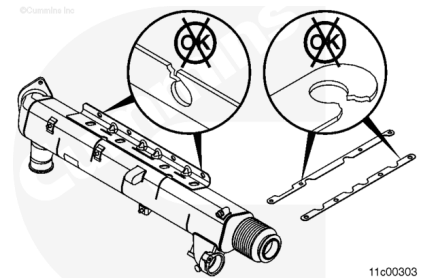
Replace hose clamp(s) if cracks or other damage is found.

Inspect hose(s) for cracks or other damage.

Replace hose(s) if cracks or other damage is found.

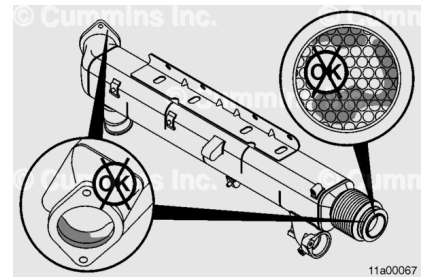


Inspect EGR cooler and the EGR cooler clamping plates for cracks or other damage. Replace the EGR cooler or EGR cooler clamping plates if cracks or other damage is found.



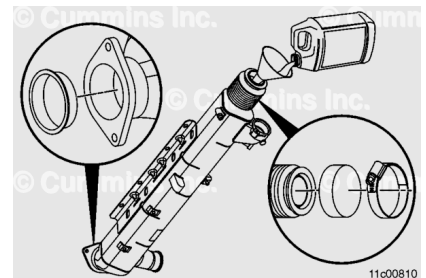
Inspect exhaust gas inlet, exhaust gas outlet, and exhaust gas tubes for coolant deposits and note their location for reference during the pressure test procedure.

Note : Reference the following service bulletin for more information regarding inspection of the EGR cooler. Reference the EGR Cooler Reuse Guidelines for X Series Engines, Bulletin 2883594.



⚠ WARNING ⚠

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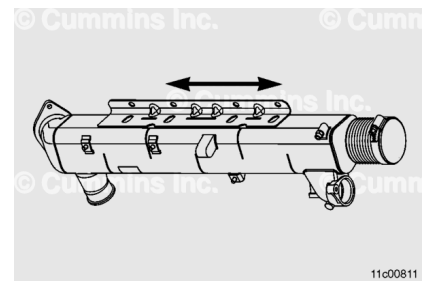
Cap outlet end of the EGR cooler with the plastic caps supplied in the EGR Cooler Leak Test Kit, Part Number 4918655 or 2892437.

Completely fill the EGR cooler with safety solvent.

Cap gas inlet end of the EGR cooler with the plastic cap supplied in the EGR Cooler Leak Test Kit, Part Number 4918655 or 2892437.

⚠ WARNING ⚠

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Lay EGR cooler on its side.

Allow EGR cooler to soak for approximately 20 minutes.

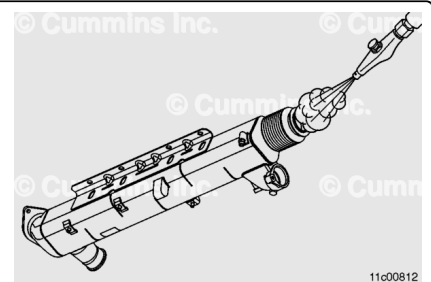
Drain approximately 25 percent of the liquid from the cooler and recap.

Properly dispose of the liquid.

Shake cooler by hand, from end-to-end, for approximately 30 seconds.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

Do not use chlorinated water to clean or rinse the EGR cooler. Chlorides are very corrosive to the stainless alloys internal and external to the EGR cooler

Drain remaining safety solvent from the EGR cooler and properly dispose of it.

Use compressed air to dry the inside of the EGR cooler and remove any loose debris or particles.

The EGR cooler can be cleaned again, if necessary.

Cap outlet end of the EGR cooler with the plastic cap supplied in the EGR Cooler Leak Test Kit, Part Number 4918655 or 2892437.

Completely fill the EGR cooler with water.

Cap gas inlet end of the EGR cooler with the plastic cap supplied in the EGR Cooler Leak Test Kit, Part Number 4918655 or 2892437.

Shake cooler by hand, from end-to-end, for approximately 30 seconds.

Drain water from the EGR cooler and properly dispose of it.

Use compressed air to dry the inside of the EGR cooler.

Clean exhaust manifold with safety solvent, where the EGR cooler connection attaches to the exhaust manifold. Refer to Procedure 011-007 in Section 11.

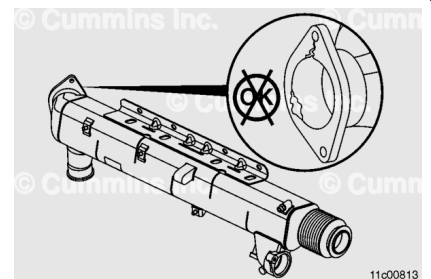
(/qs3/pubsys2/xml/en/procedures/10/10-011-007-tr.html)

Inspect exhaust manifold where the EGR cooler connection attaches to the exhaust manifold. Replace the exhaust manifold if cracks or other damage is found. Refer to Procedure 011-007 in Section 11.

(/qs3/pubsys2/xml/en/procedures/10/10-011-007-tr.html)

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris or dirt can cause personal injury.



Clean EGR cooler exhaust gas outlet with safety solvent.

Dry with compressed air.

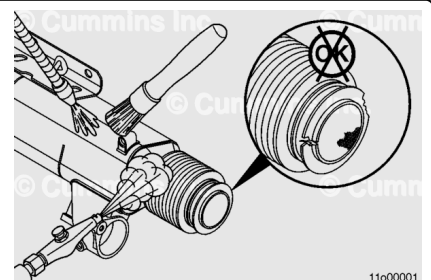
Inspect EGR cooler exhaust gas outlet for signs of fretting, cracks, or other damage.

Replace EGR cooler if damage is found. Refer to Procedure 011-025 in Section 11.

(/qs3/pubsys2/xml/en/procedures/10/10-011-025-tr.html)

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Clean EGR cooler bellows with safety solvent and dry with compressed air.

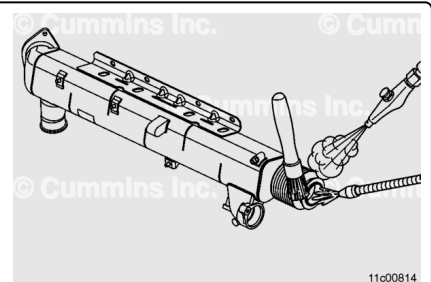
Inspect EGR cooler bellows for cracks, dents, bending, or other damage.

Replace EGR cooler if cracks, dents, bending, or other damage is found.

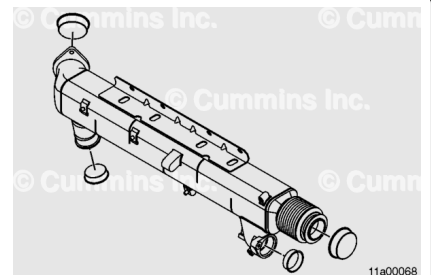
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Clean mounting surfaces with safety solvent to remove any coolant deposits or debris.

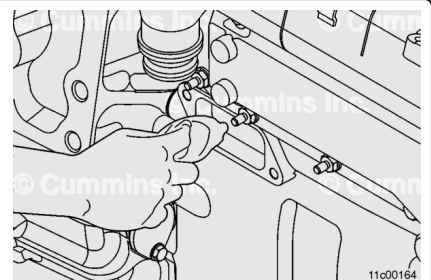


After parts are clean and dry, cover open connection points with protective caps from the ISX Clean Care Kit, Part Number 4919403, or with heavy tape, if needed.



Clean opening in the block where the EGR cooler coolant return elbow connection is installed to remove any coolant deposits or debris. Otherwise, the press-in-place seal joint can leak.

Clean EGR cooler coolant supply and return jumper tubes to remove any coolant deposits or debris.



Pressure Test

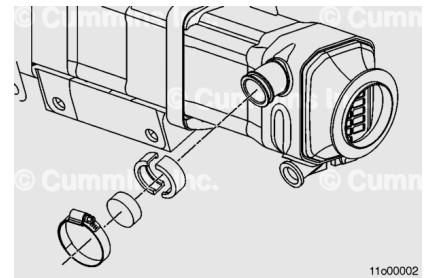
Automotive with CM870

Note : This test is performed with the EGR cooler removed from the engine.

Clean EGR cooler using the steps found in the Clean and Inspect for Reuse section of this procedure, if **not** already completed.

Plug large coolant supply port of the EGR cooler with cap, Part Number 3164738, and collar, Part Number 3164986, from the EGR Cooler Leak Test Kit, Part Number 3164225.

Secure hose clamp, Part Number 3043487.

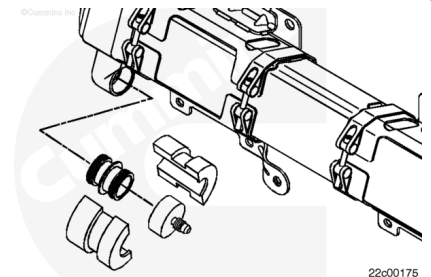


Note : Plain coupling, Part Number 3683068, and o-ring, Part Number 3049221, from EGR cooler, are used in this step.

Insert plain coupling, Part Number 3683068, in the EGR cooler coolant return port.

Install cap, Part Number 3164737, and collar, Part Number 3164736, from the EGR Cooler Leak Test Kit, Part Number 3164225, over the plain coupling, Part Number 3683068.

Secure with hose clamp, Part Number 3056787.

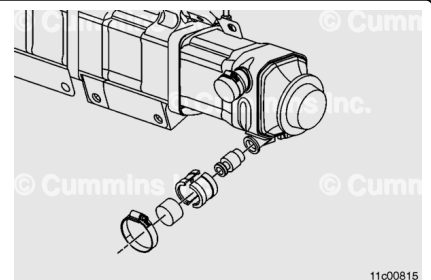


Note : Vent tube, Part Number 3682540, and the o-ring, Part Number 3682076, from the EGR cooler, are used in this step.

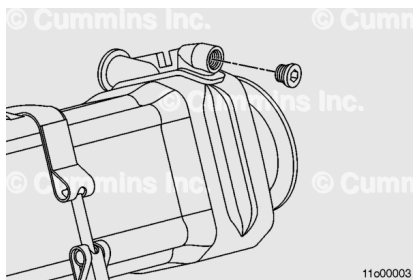
Insert vent tube, Part Number 3682540, from the EGR Cooler Leak Test Kit, Part Number 3164225, in the small EGR cooler coolant supply port.

Install the cap, Part Number 3164739, and collar, Part Number 3164985, from the EGR Cooler Leak Test Kit, Part Number 3164225, over the vent tube, Part Number 3682540.

Secure with hose clamp, Part Number 209773.

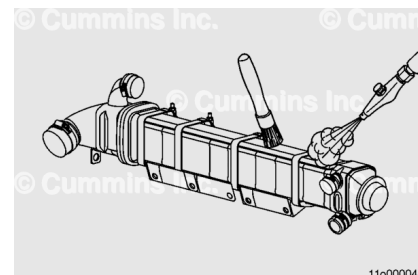


Install threaded plug, Part Number 3678923, from the EGR Cooler Leak Test Kit, Part Number 3164225, in the EGR cooler coolant vent port.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Note : The EGR cooler must be properly cleaned of soot accumulation and other debris before submerging it in the test tank.

Note : Air Pressure Regulator Kit, Part Number 3164231, or equivalent, is suitable for use during this step.

Install air pressure regulator, Part Number 3164231, or equivalent, onto the plug, Part Number 3164737, in the EGR cooler coolant return port.

Make sure the air pressure regulator is closed and connect the compressed air to the pressure regulator.

Completely submerge the EGR cooler in room temperature water. Make sure to submerge the cooler vertically, with the outlet end down, so air can **not** be left in the open (gas) side of the cooler.

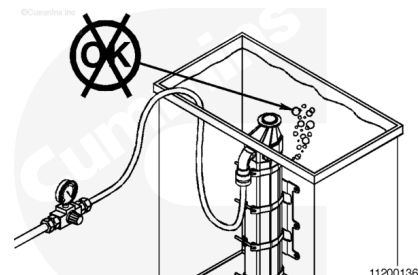
Apply 206 kPa [30psi] of air pressure to the cooler.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Inspect for bubbles escaping from uncapped gas ports of the EGR cooler or from the exterior shell of the cooler.

Verify that bubbles are **not** a result of loose fittings or trapped air.



Note : Bubbles coming from a damaged EGR cooler can be very small and/or not in a steady stream.

If bubbles are **not** observed, the EGR cooler is usable.

Make sure the pressure regulator is closed.

Disconnect the air pressure from the EGR cooler assembly.

Carefully vent the trapped air from the EGR cooler assembly.

Remove the parts supplied in the EGR Cooler Leak Test Kit,
Part Number 3164225.

If EGR cooler is **not** usable, inspect the turbocharger turbine housing and rear EGR connection tube for any signs of coolant deposits. Coolant deposits can be identified as dried white deposits coating the inside of the turbine housing or EGR connection tube.

If coolant deposits are found in the turbocharger turbine housing, inspect the turbocharger for progressive damage. Refer to Procedure 010-033 in Section 10.

(/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)

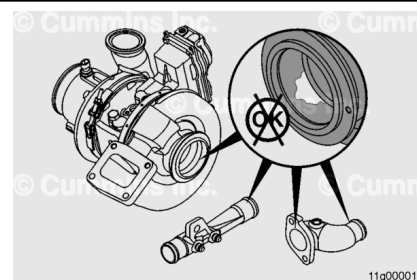
If coolant deposits are found in the EGR connection tube, inspect the EGR connection tube(s) and EGR mixer for reuse. Refer to Procedure 011-025 in Section 11.

(/qs3/pubsys2/xml/en/procedures/10/10-011-025-tr.html)

If the EGR cooler is being replaced due to damage that caused an internal coolant leak:

- Replace EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11.
(/qs3/pubsys2/xml/en/procedures/10/10-011-026-tr.html)
- Check for EGR differential pressure sensor fault codes. If fault codes are found, follow the appropriate troubleshooting tree, and replace the EGR differential pressure sensor, if required. Use the following procedure in the Signature and ISX CM870 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4021334. Refer to Procedure 019-370 in Section 19.
(/qs3/pubsys2/xml/en/procedures/70/70-019-370.html)

Note : If damage resulted in a hydraulic lock and the coolant level reached the intake manifold, it may be necessary to clean the charge air cooler. See equipment manufacturer service information.



Note : If damage resulted in a hydraulic lock and the coolant level reached the intake manifold, inspect the air compressor. Refer to Procedure 012-014 in Section 12. (</qs3/pubsys2/xml/en/procedures/10/10-012-014-tr.html>)

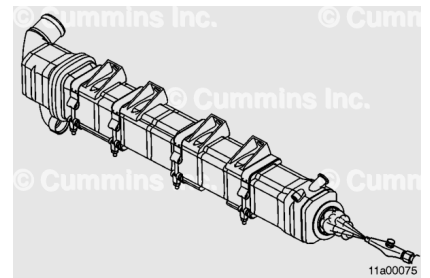
Note : If damage resulted in a hydraulic lock, it may be necessary to remove the injectors and evacuate coolant from the cylinder(s). Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/10/10-006-026-tr.html>)

⚠ WARNING ⚠

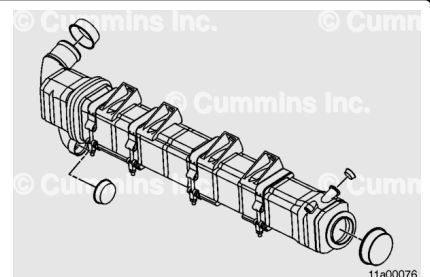
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Drain all the water from the gas side of the EGR cooler before reuse.

Dry cooler with compressed air.



After testing is completed and parts are dry, cover open connection points with protective caps from the ISX Clean Care Kit, Part Number 4919403, or with heavy tape, if needed.



Automotive With CM871

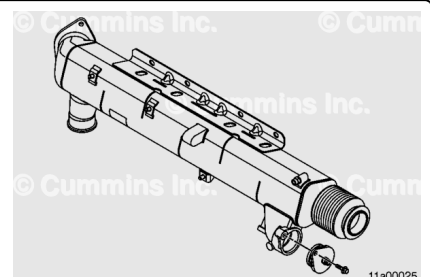
Note : This test is performed with the EGR cooler removed from the engine.

Clean EGR cooler using the steps found in the Clean and Inspect for Reuse section of this procedure, if **not** already completed.

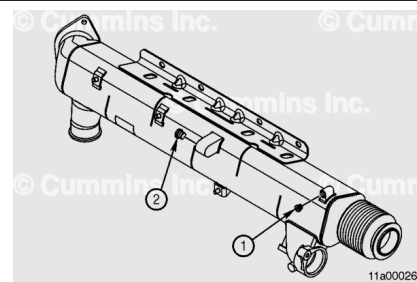
Install plug, Part Number 4918689, with the o-ring, Part Number 3683814, supplied in the EGR Cooler Leak Test Kit, Part Number 4918655 or 2892437, in the coolant supply port of the EGR cooler.

Secure plug with a M8 x 1.25 capscrew.

Torque Value: 21 n•m [186 in-lb]



Install threaded plug (1), Part Number 4002056, and the o-ring, Part Number 3348319, from the EGR Cooler Leak Test Kit, Part Number 4918655 or 2892437, in the EGR cooler coolant vent port.



Install threaded plug (2), Part Number 3089567, and the o-ring, Part Number 3679139, from the EGR Cooler Leak Test Kit, Part Number 4918655 or 2892437, in the aftertreatment fuel injector coolant supply port.

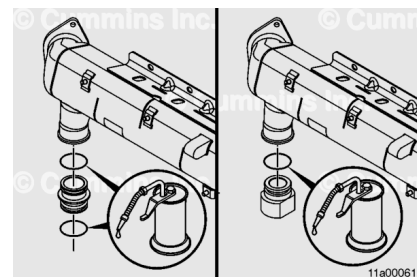
Locate EGR cooler coolant return coupling, Part Number 3683586.

Inspect o-rings, Part Number 3683814, for cracks, cuts, or other damage. Replace if necessary.

Lubricate o-rings with clean engine coolant, soapy water, or vegetable oil to aid in installation.

Install EGR cooler coolant return coupling into the EGR cooler coolant return port.

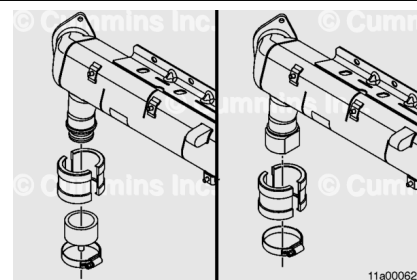
Note : If leak test plug, Part Number 2892396, is available, install it into the EGR cooler instead of the EGR cooler coolant return coupling, Part Number 3683586.



Install collar, Part Number 4918446, and the cap, Part Number 4918444, over the EGR cooler coolant return coupling and EGR cooler coolant return port.

Note : If leak test plug, Part Number 2892396, is available, the cap, Part Number 4918444, will not be needed for this step.

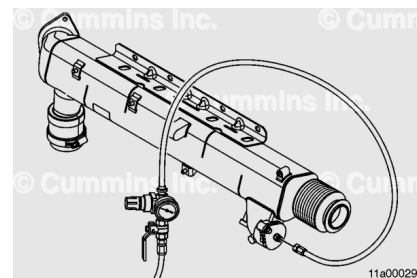
Secure with hose clamp, Part Number 3026396, from the EGR Cooler Leak Test Kit, Part Number 4918655 or 2892437.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Note : Air pressure regulator kit, Part Number 3164231, or equivalent, is suitable for use during this step.



Install air pressure regulator, Part Number 3164231, or equivalent, on the plug, Part Number 4918689, in the EGR cooler coolant supply port.

Install a ball-type shutoff valve onto the supply side of the air pressure regulator, to allow the air supply to be shut off during testing.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Note : The test must be performed when the EGR cooler has reached ambient temperature. Do not apply air pressure to a hot EGR cooler.

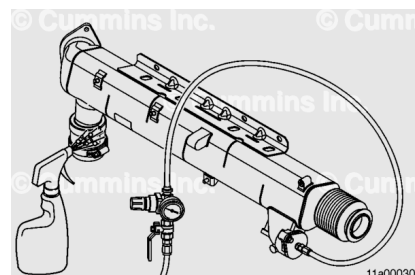
Make sure air pressure regulator and ball-type shutoff valve are closed and connect a compressed air supply to the pressure regulator.

Apply 620 kPa [90 psi] of air pressure to the EGR cooler.

Inspect for air escaping from the EGR cooler or pressure test assembly.

Check for leaks by spraying the exterior of the EGR cooler and all lines and fittings with a mixture of mild soap and water. Repair leaking fittings or lines.

Replace EGR cooler if external leaks, caused by cracks or other damage, are found on EGR cooler.



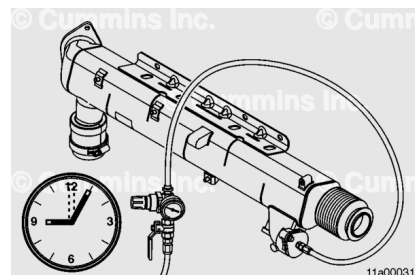
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Shut air supply off to the air pressure regulator, Part Number 3164231, using the ball valve.

Disconnect compressed air supply from the air pressure regulator.

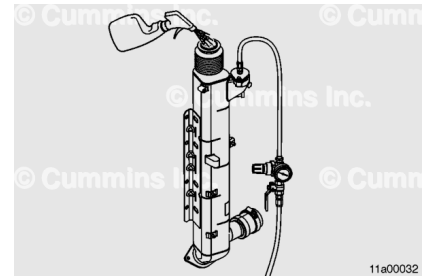
Record time the compressed air supply was shut off and removed from the air pressure regulator.



Record pressure shown on the air pressure regulator gauge.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

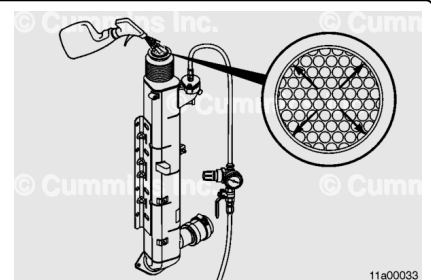


Place EGR cooler in a vertical position with the exhaust gas inlet bellows pointing upward.

Spray exhaust inlet bulkhead and gas tubes with a mixture of mild soap and water.

Gently flex the exhaust gas inlet bellows so that the inside corners of the bulkhead can be sprayed as well.

Note : It is important to wet the entire bulkhead with the mixture of mild soap and water. The highest percentage of leaks occur in the inside corners of the EGR cooler.



Allow EGR cooler to remain pressurized for 15 minutes.

Gently flex bellows and inspect inside corners of bulkhead and gas tubes for continuous formation of bubbles.

Measure and record air pressure.

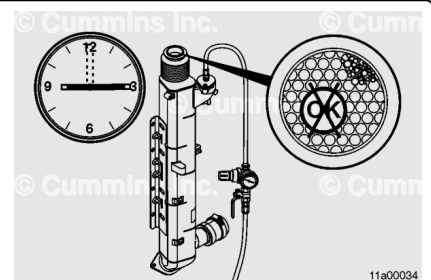
If the measured pressure loss in the EGR cooler is more than 103 kPa [15 psi] in 15 minutes, or if there is a continuous formation of bubbles near the gas tubes and bulkhead, the EGR cooler is **not** usable

Note : The pressure test is meant to identify large leaks and the inspection with mild soap and water is meant to identify small leaks. Do not use small losses observed during the pressure test **only** to evaluate the EGR cooler for reuse, as they are often associated with issues with the leak test equipment.

Note : Stationary bubbles that form on the soot deposits or in the gas tubes are **not** an indication of a leak.

Note : The continuous formation of bubbles is an indication that the EGR cooler is **not** usable.

Note : If EGR cooler is **not** usable based on a loss of pressure, carefully inspect the air pressure regulator, air



lines, ball valves, o-rings, and other components for leaks. A mixture of mild soap and water may be sprayed on the assembly to aid in leak detection.

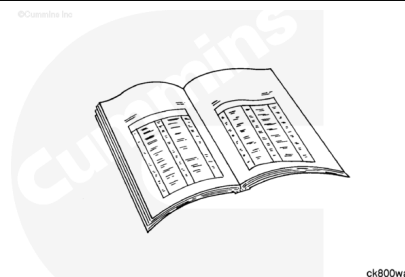
Note : If EGR cooler is **not** usable based on a loss of pressure, it can be necessary to evaluate the EGR cooler pressure testing assembly for leaks before replacing the EGR cooler. Assorted fittings or a known good EGR cooler can be used for this test.

Make sure pressure regulator is closed.

Disconnect air pressure from the EGR cooler assembly.

Carefully vent trapped air from EGR cooler assembly.

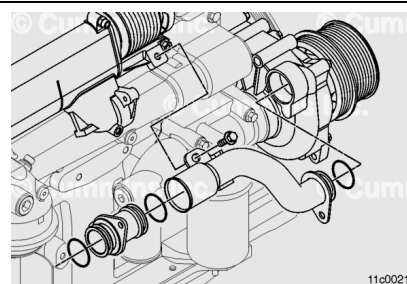
Remove components installed from EGR Cooler Leak Test Kit, Part Number 4918655 or 2892437.



If EGR cooler is usable, inspect the EGR cooler coolant lines, couplings, o-rings, and grommets for cracks or other damage.

Replace necessary components if cracks or other damage is found. Refer to Procedure 011-031 in Section 11.

(</qs3/pubsys2/xml/en/procedures/10/10-011-031-tr.html>)



If EGR cooler is **not** usable, inspect the turbocharger turbine housing and rear EGR connection tube for any signs of coolant deposits.

Coolant deposits can be identified as dried white deposits coating the inside of the turbine housing.

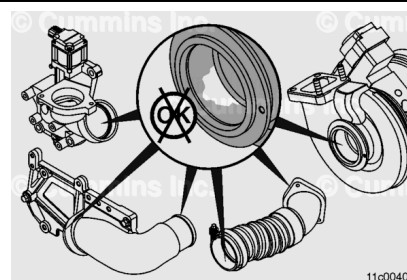
If coolant deposits are found in the turbocharger turbine housing, check for turbocharger related fault codes.

If turbocharger related fault codes are found, inspect the turbocharger. Refer to Procedure 010-033 in Section 10.
(</qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html>)

If coolant deposits are found in the rear EGR connection tube, inspect the EGR valve for progressive damage. Refer to Procedure 011-022 in Section 11.

(</qs3/pubsys2/xml/en/procedures/10/10-011-022-tr.html>)

If coolant deposits are found in the EGR valve, inspect the EGR venturi and front EGR connection tubes for progressive damage. Refer to Procedure 011-025 in Section 11.



(/qs3/pubsys2/xml/en/procedures/10/10-011-025-tr.html)

If EGR cooler is being replaced due to damage that caused an internal coolant leak:

- Clean aftertreatment fuel injector, regardless if coolant deposits are present or **not**. Refer to Procedure 011-042 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-042-tr.html)
- If Fault Code 555 and/or 556 is present, replace the crankcase breather element. Refer to Procedure 003-019 in Section 3. (/qs3/pubsys2/xml/en/procedures/101/101-003-019-tr.html)
- Inspect crankcase pressure sensor. Use the following procedure in the ISX CM871 and ISM CM876 Electronic Control System Troubleshooting Manual, Bulletin 4021560. Refer to Procedure 019-445 in Section 19. (/qs3/pubsys2/xml/en/procedures/00/00-019-445.html)
- Inspect EGR differential pressure sensor tubes for coolant. If coolant is found, replace the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11. (/qs3/pubsys2/xml/en/procedures/10/10-011-026-tr.html)
- Check for EGR differential pressure sensor fault codes. If fault codes are found, follow the appropriate troubleshooting tree, and replace the EGR differential pressure sensor if, required. Use the following procedure in the ISX CM871 and ISM CM876 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4021560. Refer to Procedure 019-370 in Section 19. (/qs3/pubsys2/xml/en/procedures/16/16-019-370.html)
- If coolant entered exhaust system, aftertreatment system can be recovered. Refer to Procedure 014-013 in Section 14. (/qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html)

Note : If damage resulted in a hydraulic lock and the coolant level reached the intake manifold, it may be necessary to clean the charge air cooler. See equipment manufacturer service information.

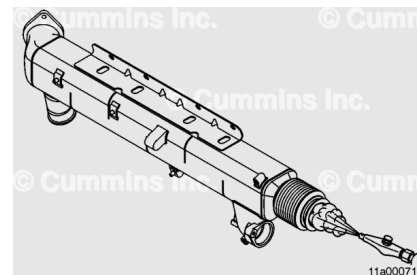
Note : If damage resulted in a hydraulic lock and/or the coolant level reached the intake manifold, inspect the air compressor air inlet and outlet tubes for signs of coolant. If coolant is found, clean and inspect the air compressor head. Refer to Procedure 012-007 in Section 12.

(/qs3/pubsys2/xml/en/procedures/101/101-012-007-tr.html)

Note : If damage resulted in a hydraulic lock, it may be necessary to remove the injectors and evacuate coolant from the cylinder(s). Refer to Procedure 006-026 in Section 6.
(/qs3/pubsys2/xml/en/procedures/10/10-006-026-tr.html)

⚠ WARNING ⚠

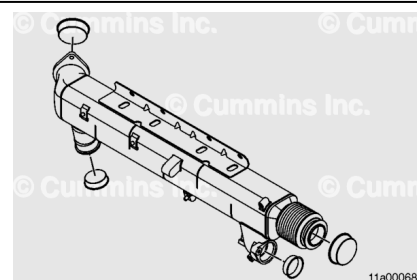
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Drain all the water from the gas side of the EGR cooler before reuse.

Dry cooler with compressed air.

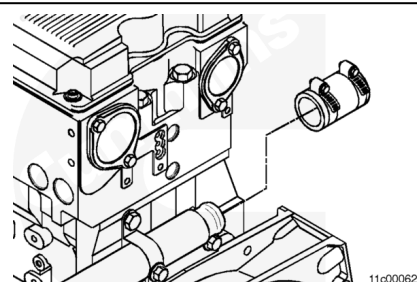
After testing is completed and parts are dry, cover open connection points with protective caps from the ISX Clean Care Kit, Part Number 4919403, or with heavy tape, if needed.



Install

Automotive with CM870

Install hose onto the rear EGR connection tube and install the hose clamps.



Remove any protective caps or heavy tape.

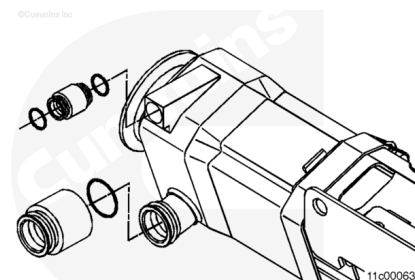
Install new o-rings on the coolant supply tube and vent tube on the EGR cooler.

Use clean engine coolant, soapy water, or vegetable oil to lubricate the o-rings for easier installation.

Install coolant supply tube and the vent tube on the EGR cooler.

Align coolant supply and vent tubes to the corresponding openings in the block.

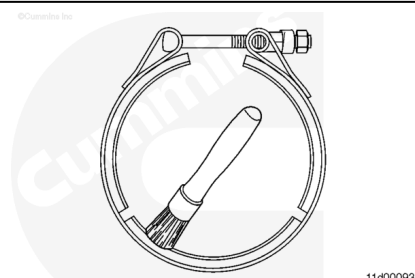
Install EGR cooler on the studs in the lubricating oil cooler assembly.



Install a new gasket on the pilot end of the EGR cooler, where it connects to the EGR cooler connection.

Apply a film of high temperature anti-seize compound to the inside of the V-band clamp to make sure of proper loading on the clamp.

Install a V-band clamp.



Torque Value: 17 n•m [150 in-lb]

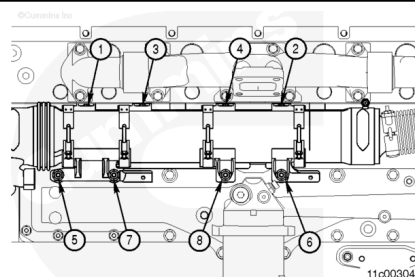
Install and hand-tighten the four mounting nuts and capscrews. Tighten the capscrews and nuts in the sequence shown.

Torque Value:

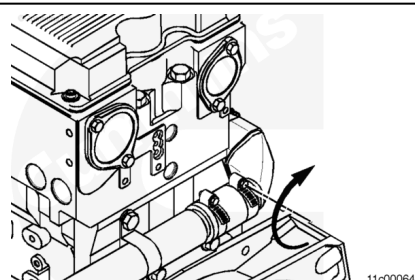
Capscrews 47 n•m [35 ft-lb]

Torque Value:

Nuts 23 n•m [204 in-lb]



Connect EGR cooler to the hose on the rear EGR connection tube.



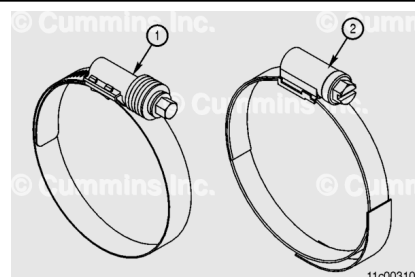
Tighten hose clamps.

Torque Value:

Item 1: 7 n•m [62 in-lb]

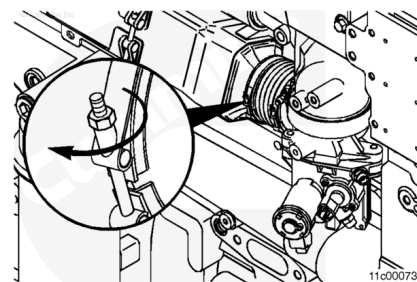
Torque Value:

Item 2: 5 n•m [44 in-lb]



Tighten V-band clamp between the EGR cooler connection and the EGR cooler.

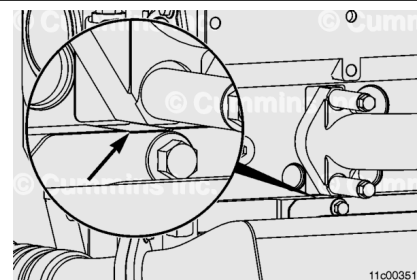
Torque Value: 17 n·m [150 in-lb]



Automotive With CM871

⚠ CAUTION ⚠

Keep all protective caps on the EGR cooler in place while grinding clearance in the upper mounting bracket. Failure to do so could cause progressive damage to the engine once the EGR cooler is installed. It is important to prevent any debris from entering the cooler.



Inspect for an interference fit between the upper EGR cooler mounting bracket and the lower portion of the exhaust manifold flange. If this interference is present, grind a clearance on the upper EGR cooler mounting bracket in the area shown in the illustration.

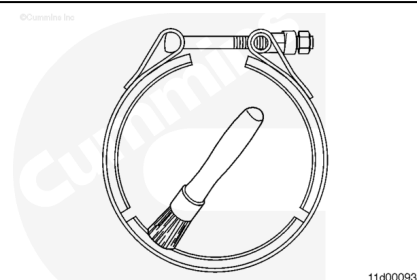
Remove any protective caps or heavy tape.

Apply a film of high temperature anti-seize compound to the inside of the V-band clamp to make sure of proper loading on the clamp.

Install a V-band clamp onto the exhaust manifold front section.

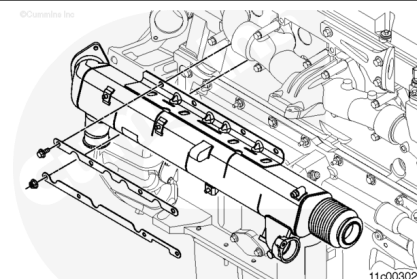
Install a new gasket on the pilot end of the EGR cooler where it connects to the exhaust manifold front section.

Install EGR cooler on the studs in the lubricating oil cooler assembly.



Install EGR cooler bracket clamping plates.

Note : Some EGR coolers use washers instead of clamping plates.



Install four EGR cooler mounting capscrews at the top of the EGR cooler mounting bracket.

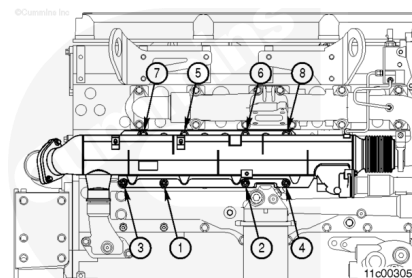
Install and hand-tighten the four mounting nuts. Tighten the capscrews and nuts on the bottom EGR cooler mounting bracket in the sequence shown.

Torque Value:

Capscrews 47 n•m [35 ft-lb]

Torque Value:

Nuts 23 n•m [204 in-lb]

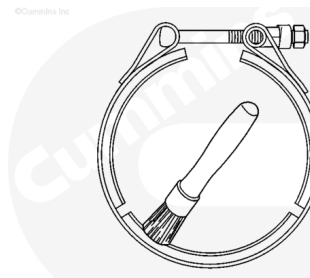


⚠ CAUTION ⚠

Do not use air tools to remove or install the nut on the V-band clamp. Use of these tools can seriously damage the threads or the bolt and cause the clamp to not be able to be reused.

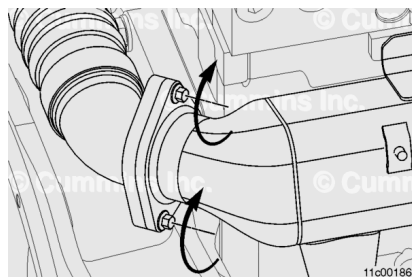
Position V-band clamp onto the EGR cooler inlet connection.

Torque Value: 17 n•m [150 in-lb]



Tighten two retaining capscrews on the rear EGR connection tube.

Torque Value: 20 n•m [177 in-lb]

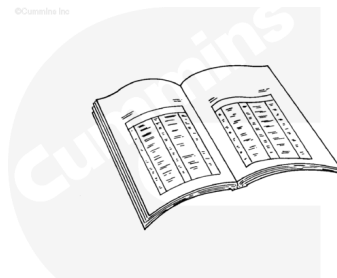


Finishing Steps

Automotive with CM870

Note : If the EGR cooler is being replaced due to an internal coolant leak, the EGR differential pressure tubes must be replaced. Refer to Procedure 011-026 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-026.html>)

- Install EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-031-tr.html>)



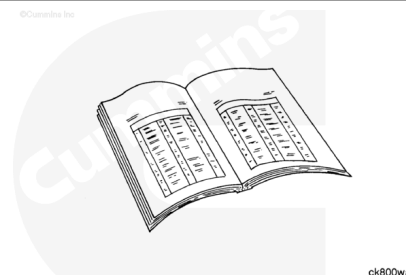
- Install exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-027-tr.html>)
- Install turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html>)
- Remove protective caps used to cover the charge air cooler pipe connection points.
- Install charge air cooler pipe. See equipment manufacturer service information.
- Remove the protective caps used to cover the air intake pipe to the turbocharger
- Install air intake pipe to the turbocharger. See equipment manufacturer service information.
- Fill engine cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>)
- Start engine, check for leaks, and verify proper operation.

Automotive With CM871

Note : If the exhaust gas recirculation (EGR) cooler is being replaced due to a malfunction that caused an internal coolant leak and Fault Code 555 and/or 556 is present, the crankcase breather system must be inspected. Maintenance-free crankcase breather inserts must be cleaned and inspected for reuse. Crankcase breather elements with filtration media must be replaced. Refer to Procedure 003-019 in Section 3. (</qs3/pubsys2/xml/en/procedures/10/10-003-019.html>)

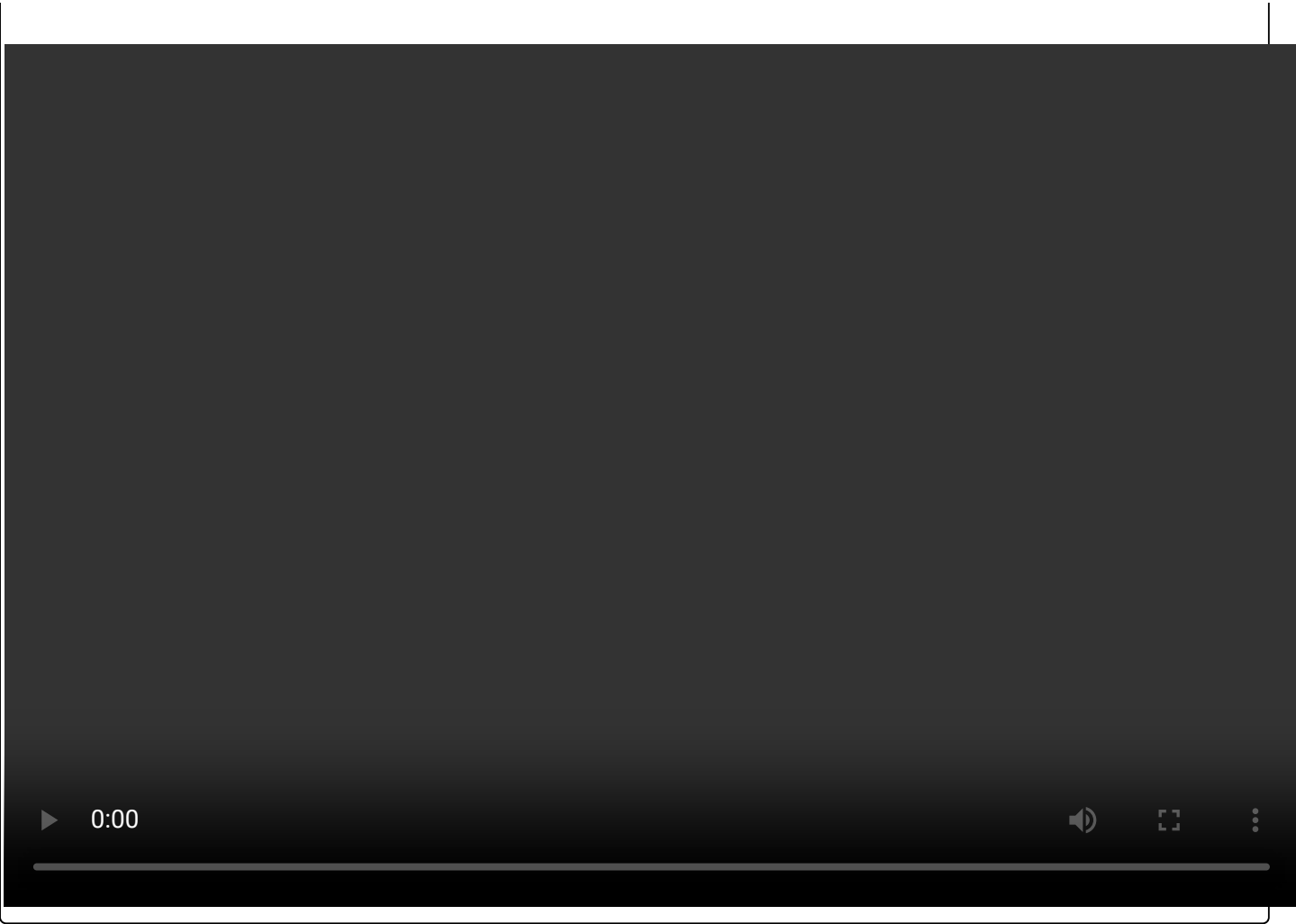
Note : If the EGR cooler is being replaced due to damage that caused an internal coolant leak, the EGR differential pressure sensor tubes are to be inspected for coolant. If coolant is found, replace the EGR differential pressure tubes. Refer to Procedure 011-026 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-026.html>)

Note : If damage resulted in oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system must be inspected. Refer to the Aftertreatment Diesel Oxidation Catalyst and Aftertreatment Diesel Particulate Filter Reuse Guidelines, Bulletin 4021600 (</qs3/pubsys2/xml/en/bulletin/4021600.html>).



Note : If damage resulted in coolant entering the exhaust system, the aftertreatment system can be recovered. Refer to Procedure 014-013 in Section 14. (</qs3/pubsys2/xml/en/procedures/101/101-014-013.html>)

- Install EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-031-tr.html>)
- Install turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html>)
- Install aftertreatment adapter pipe and aftertreatment injector lines. Refer to Procedure 011-043 (</qs3/pubsys2/xml/en/procedures/101/101-011-043-tr.html>) and Refer to Procedure 011-051 in Section 11. (</qs3/pubsys2/xml/en/procedures/101/101-011-051-tr.html>)
- Remove protective caps used to cover the charge air cooler pipe connection points.
- Install charge air cooler pipe. See equipment manufacturer service information.
- Remove protective caps used to cover the air intake pipe connection points.
- Install air intake pipe to the turbocharger. See equipment manufacturer service information.
- Fill engine cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>)
- Start engine, check for leaks, and verify proper operation.



Last Modified: 27-Aug-2018
